

Interfacing IR Sensor, Ultrasonic Sensor, and DHT11 Sensor using Arduino Uno.

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Objective

To study and implement the working of different sensors (IR, Ultrasonic, and DHT11) using Arduino Uno and analyze their outputs through the Serial Monitor.

Components Required

- Arduino Uno
- IR Sensor Module
- Ultrasonic Sensor (HC-SR04)
- DHT11 Temperature & Humidity Sensor
- Jumper Wires
- USB Cable

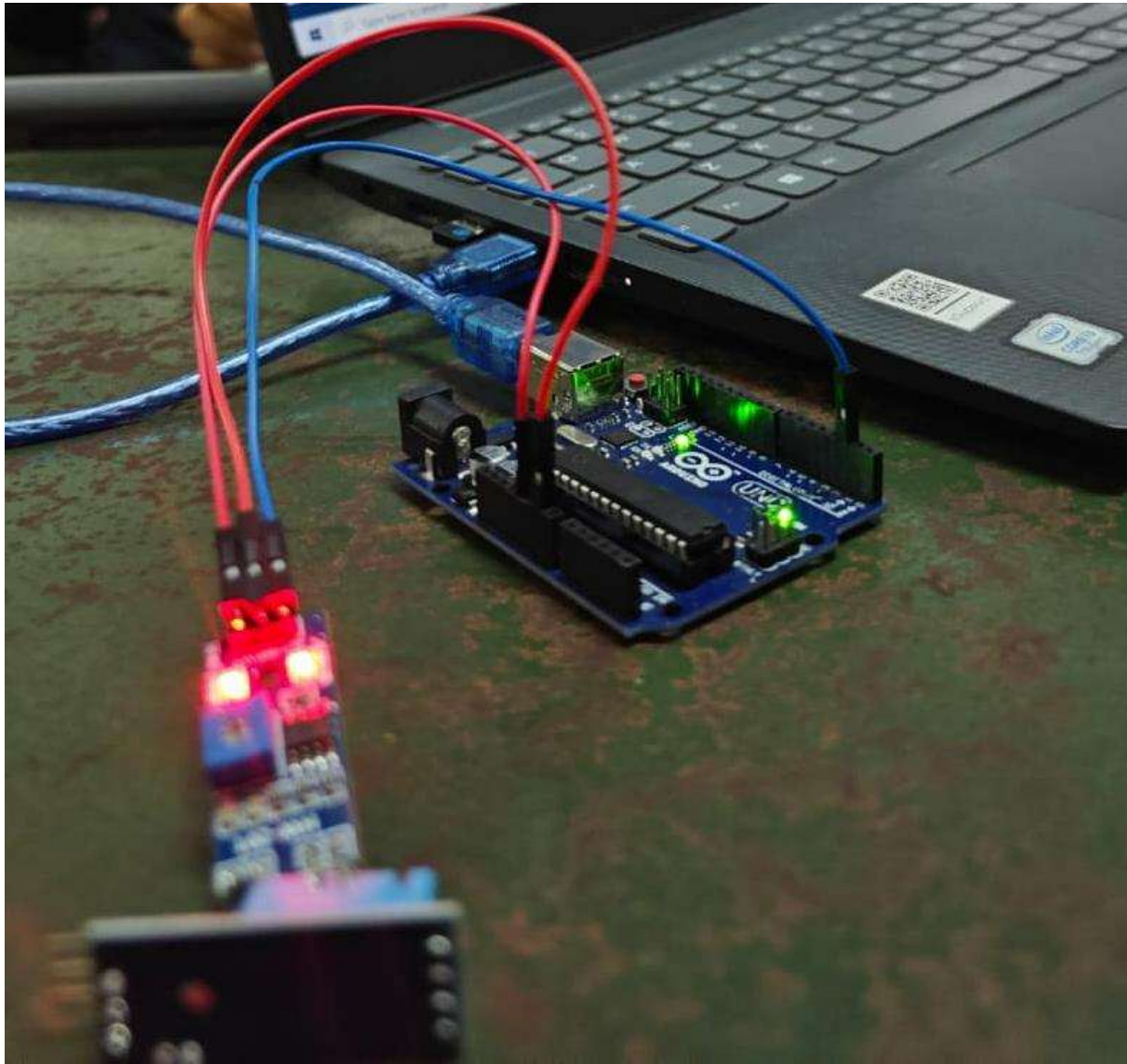
1. IR SENSOR INTERFACING

Principle

The IR sensor detects objects based on infrared reflection. When an object comes close, the IR light reflects back and is detected.

Pin Configuration

IR Sensor Pin	Arduino Uno
VCC	5V
GND	GND
OUT	Pin 2



Code

```
int sensorPin = 2;

void setup() {
  pinMode(sensorPin, INPUT);
  Serial.begin(9600);
}

void loop() {
  int value = digitalRead(sensorPin);

  if (value == LOW) {
    Serial.println("Object Detected");
  } else {
    Serial.println("No Object");
  }

  delay(500);
}
```



Working

- Sensor outputs **LOW** when object is detected
- Arduino reads signal and prints result
- Continuous monitoring using loop

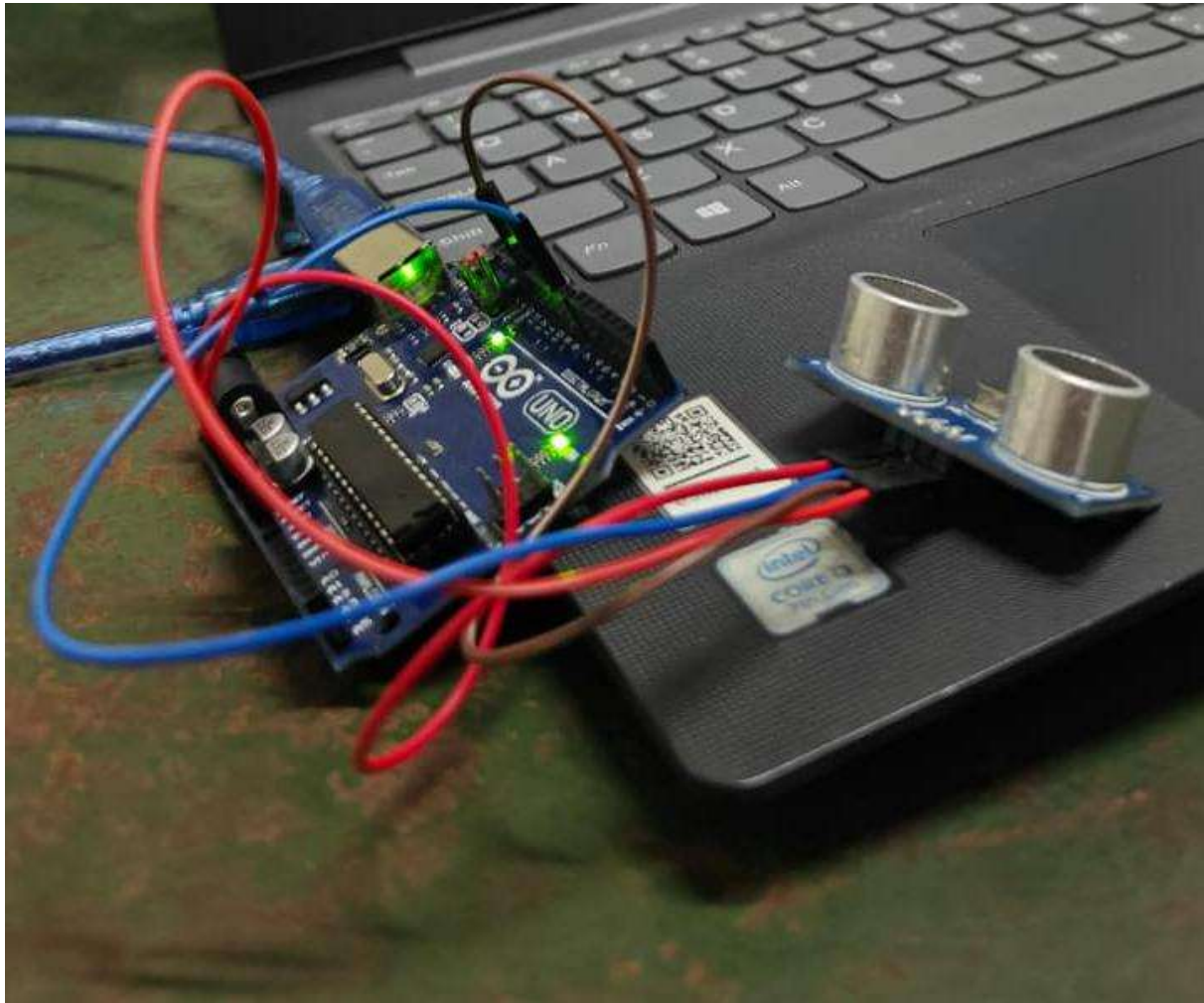
2. ULTRASONIC SENSOR INTERFACING

Principle

The ultrasonic sensor measures distance using sound waves. It calculates the time taken for the echo to return.

Pin Configuration

Sensor Pin	Arduino Uno
VCC	5V
GND	GND
TRIG	Pin 9
ECHO	Pin 10



Code

```
int trigPin = 9;
int echoPin = 10;

long duration;
int distance;

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  Serial.begin(9600);
}

void loop() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  duration = pulseIn(echoPin, HIGH, 30000);
```

```
distance = duration * 0.0343 / 2;
```

```
Serial.print("Distance: ");
```

```
Serial.print(distance);
```

```
Serial.println(" cm");
```

```
delay(500);
```

```
}
```

Working

- Trigger sends ultrasonic wave
- Echo receives reflected wave
- Time is measured
- Distance calculated using formula:
Distance = (Speed × Time) / 2

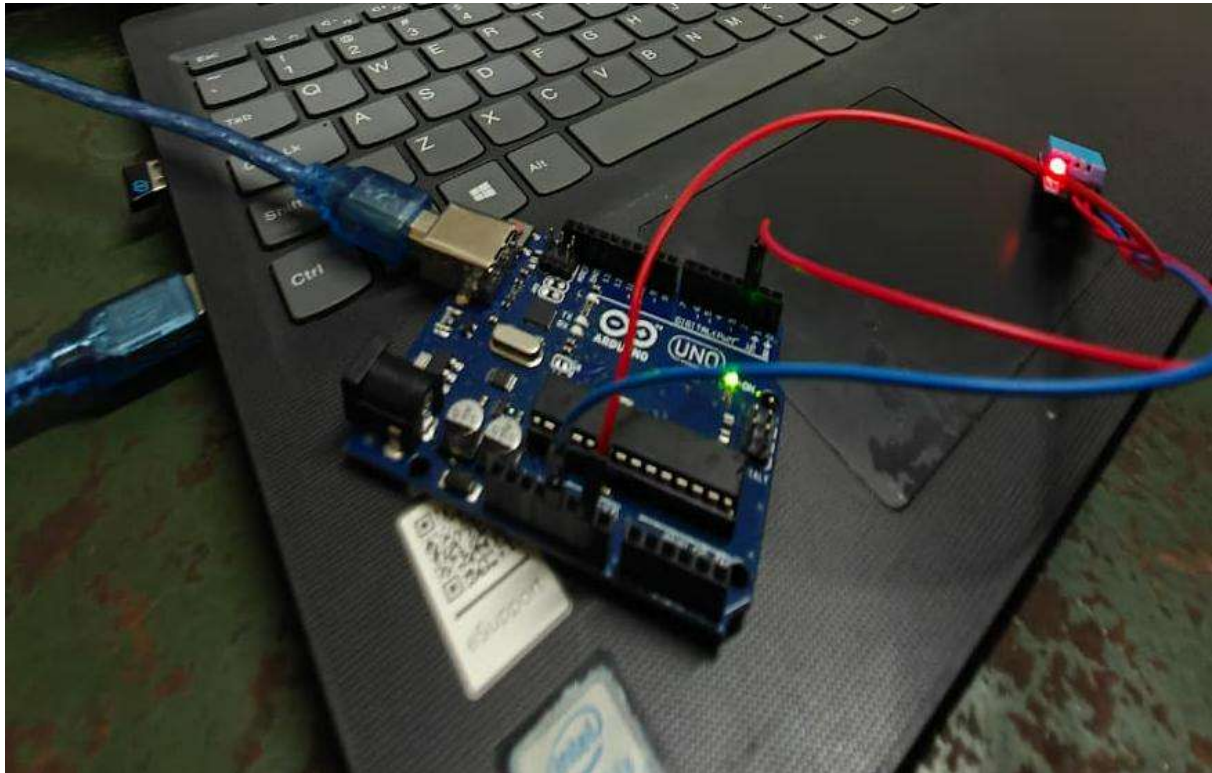
3. DHT11 SENSOR INTERFACING

Principle

The DHT11 sensor measures temperature and humidity using a digital signal output.

Pin Configuration

DHT11 Pin	Arduino Uno
VCC	5V
GND	GND
DATA	Pin 2



Code

```
#include <DHT.h>

#define DHTPIN 2
#define DHTTYPE DHT11

DHT dht(DHTPIN, DHTTYPE);

void setup() {
  Serial.begin(9600);
  dht.begin();
}

void loop() {
  float temperature = dht.readTemperature();
  float humidity = dht.readHumidity();

  if (isnan(temperature) || isnan(humidity)) {
    Serial.println("Failed to read from DHT sensor!");
    return;
  }

  Serial.print("Temp: ");
  Serial.print(temperature);
  Serial.print(" °C | Humidity: ");
  Serial.print(humidity);
  Serial.println(" %");
}
```



```
delay(2000);
```

```
}
```

Working

- Sensor sends digital data
- Arduino reads temperature & humidity
- Displays values on Serial Monitor
- Delay ensures stable readings

Results

- IR sensor successfully detected objects
- Ultrasonic sensor measured distance (in cm)
- DHT11 provided temperature and humidity readings

Applications

- Smart dustbin system
- Obstacle detection
- Environmental monitoring
- Automation systems

Conclusion

The experiment successfully demonstrated the interfacing of multiple sensors with Arduino Uno. Each sensor operates on a different principle (infrared, ultrasonic, and digital sensing), providing a strong foundation for building real-time embedded systems.